

# 1 Part 20: Bayesian Estimation

## 2 Summary

3 To this point, our discussion has focused on the **frequentist** or **classical**  
4 approach to estimation and hypothesis testing. In that framework, consid-  
5 eration is given to how procedures perform in repeated use. Indeed, the  
6 performance of estimators can be assessed without using any actual data.  
7 Some find this troubling.

8 An alternative philosophy is the **Bayesian** approach to estimation. The  
9 central characteristic is that uncertainty in parameters is quantified using  
10 probability distributions.

11 As the name implies, Bayes' Rule plays a central role.

1 First, recall Bayes' Rule: If  $B_1, B_2, \dots, B_m$  are a partition of  $\Omega$ , then

$$2 \quad P(B_i|A) = P(A|B_i)P(B_i) \bigg/ \sum_j P(A|B_j)P(B_j)$$

3 Also recall the continuous version:

$$4 \quad f_{Y|X}(y|x) = f_{X|Y}(x|y)f_Y(y) \bigg/ \int f_{X|Y}(x|u)f_Y(u) du$$

5 Here we will use this result with the parameter  $\theta$  "playing the role" of  $Y$ :

$$6 \quad \pi(\theta|\mathbf{x}) = f(\mathbf{x}|\theta)\pi(\theta) \bigg/ \int f(\mathbf{x}|t)\pi(t) dt$$

7 The vector  $\mathbf{x}$  is observed data.

8 The density  $\pi(\theta)$  is the **prior distribution** for  $\theta$ .

9 The conditional density  $\pi(\theta|\mathbf{x})$  is the **posterior distribution** for  $\theta$ .

- 1 In the Bayesian framework, the prior reflects the someone's **opinion** re-
- 2 garding the value of  $\theta$ , **prior to** the observation of the data.
- 3 The posterior is this opinion “updated” to take the data into consideration.
- 4 *How is the prior chosen?* This can be tricky, and subjective.
- 5 The use of the Bayesian approach was long limited by computational diffi-
- 6 culties associated with calculating the required integral. With the advent of
- 7 **Markov Chain Monte Carlo (MCMC)** algorithms, these issues have largely
- 8 gone away, and the use of Bayesian methods has grown greatly.
- 9 In some relatively simple cases, analytical solutions are possible.

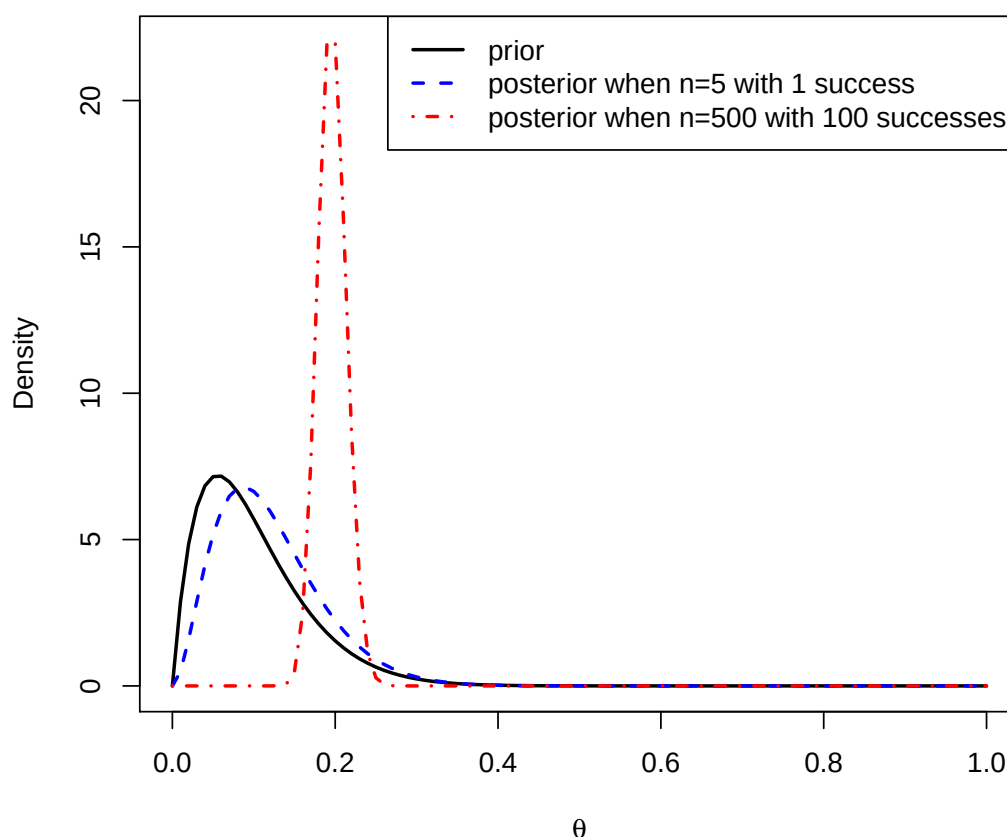
- 1 **Exercise:** Suppose that the observable data are modelled by i.i.d.
- 2 Bernoulli( $\theta$ ) random variables  $X_1, X_2, \dots, X_n$ . You quantify your prior
- 3 beliefs regarding  $\theta$  by using the Beta( $\alpha, \beta$ ) distribution, for some chosen
- 4 values of  $\alpha$  and  $\beta$ . Derive the posterior for  $\theta$  in this case.

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- 1 In this example, we see that the prior and posterior are from the same  
2 family of distributions (i.e., they are both Beta). We say that “The Beta  
3 distribution is conjugate to sampling from the Bernoulli.”
- 4 Let’s consider a couple specific examples. Suppose I chose  $\alpha = 2$  and  
5  $\beta = 18$  to capture my prior opinion regarding  $\theta$ . I observe  $n = 5$  trials, and  
6  $\sum_i x_i = 1$ . The result above states that the posterior is Beta( $2+1, 18+(5-1)$ ).
- 7 If instead,  $n = 500$  and  $\sum_i = 100$ , the posterior is Beta( $102, 418$ ).
- 8 The figures on the following slide compare these cases.



- 1 **Exercise:** Comment on the shapes of the posteriors in the previous exam-  
 2 ple, as compared to the prior. What is the role of the sample size?

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- 1 The Bayesian often wants to convert the posterior into an estimate for  $\theta$ .  
 2 One way to do this is to find the maximum of the posterior, called the  
 3 **maximum a posteriori (MAP) estimator**.

- 4 In our example this comes out to be

$$\frac{\alpha + \sum x_i - 1}{\alpha + \beta + n - 4}$$

- 6 Another approach is to find the posterior mean, called the **Bayes estimator**.

- 7 In our example the Bayes estimator is

$$E(\theta|\mathbf{x}) = \frac{\alpha + \sum x_i}{\alpha + \beta + n} = \left(\frac{\sum x_i}{n}\right) \left(\frac{n}{\alpha + \beta + n}\right) + \left(\frac{\alpha}{\alpha + \beta}\right) \left(\frac{\alpha + \beta}{\alpha + \beta + n}\right)$$

- 9 In other words, it is a weighted average of the sample proportion and the  
 10 prior mean. The weight placed on the sample proportion grows with  $n$ .

- 1 **Exercise:** Return to our Bernoulli example. Choose  $\alpha = \beta = 1$  for the prior.  
2 What is the appeal of this choice? What is the Bayes estimator?

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- 1 **Exercise:** The observable data are modelled using the  $\text{Poisson}(\theta)$  distribu-  
2 tion. Use the  $\text{Gamma}(\alpha, \beta)$  distribution as the prior on  $\theta$ . Find the posterior  
3 distribution and the Bayes estimator.

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## 1 Higher Dimensional Parameters

2 The Bayesian approach can be extended to multi-dimensional parameters.  
3 The difficulties are more computational than conceptual. It is possible,  
4 for instance, to place prior distributions on covariance matrices with tools  
5 tailored to this purpose. Choosing priors can be even trickier in higher  
6 dimensions, however.

7 For example, the paper “Bayesian Portfolio Analysis” by Avramov and  
8 Zhou (*Ann. Rev. of Financial Economics*, Vol. 2) provides a nice overview of  
9 Bayesian tools for selecting portfolios. Consider Equation (8) for the un-  
10 derlying optimization problem. Note how uncertainty in the mean vector  
11  $\mu$  and covariance matrix  $V$  are taken into consideration: Fixed estimates  
12 are not plugged in, instead one integrates over the posterior for these quan-  
13 tities.